Department of Mathematics, IIT Madras

MA 2102 - Differential Equations, ODD Semester, 2026-2027

Assignment-5 : Legendre Polynomials and Functions

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1. Show that Legendre polynomial P_n of degree n satisfies the following relations:

   (a) $P_n(-x) = (-1)^n P_n(x)$,

   (b) $P_n(-1) = (-1)^n$,

   (c) $P_n'(-x) = (-1)^{n+1} P_n'(x)$,

   (d) $P_n'(1) = \dfrac{n(n+1)}{2}$ and $P_n'(-1) = (-1)^{n-1}\dfrac{n(n+1)}{2}$,

   (e) $P_{2n}'(0) = 0$ and $P_{2n+1}'(0) = (-1)^n \dfrac{(2n+1)!}{2^{2n}(n!)^2}$.

2. Show that if $f(x)$ is a polynomial with double roots at a and b, then $f''(x)$ vanishes at least twice in (a, b). This result can be generalized to P_n by using Rodrigues' formula, i.e. prove that $P_n(x)$ has n distinct roots in $(-1, 1)$.

3. Find the first three terms of the Legendre series of the following functions:

   (a) $f(x) = \begin{cases} -x, & -1 < x < 0, \\ x, & 0 \le x \le 1; \end{cases}$

   (b) $f(x) = \begin{cases} 0, & -1 \le x < 0, \\ 1, & 0 \le x \le 1; \end{cases}$

   (c) $f(x) = e^x,\ -1 \le x \le 1;$

   (d) $f(x) = 1 - |x|,\ -1 \le x \le 1;$

   (e) $f(x) = \cos \pi x,\ -1 \le x \le 1;$

   (f) $f(x) = 7x^4 - 3x + 1,\ -1 \le x \le 1.$

4. Prove the following:

   (a) $(1 - x^2)P_n'(x) = n\big[P_{n-1}(x) - xP_n(x)\big]$,

   (b) $(1 - x^2)P_n'(x) = (n+1)\big[xP_n(x) - P_{n+1}(x)\big]$,

   (c) $(2n+1)(1 - x^2)P_n'(x) = n(n+1)\big[P_{n+1}(x) - P_{n-1}(x)\big]$,

   (d) $P_7'(x) = 13P_6(x) + 9P_4(x) + 5P_2(x) + P_0(x)$,

   (e) $P_6'(x) = 11P_5(x) + 7P_3(x) + 3P_1(x)$.

5. Express the following in terms of Legendre Polynomials:

   (a) $f(x) = x^2 - 5$;

   (b) $x^3 + x + 1$;

   (c) $x^3 + 2x^2 - x - 3$.

6. Prove that there exist constants $a_0, a_1, \ldots, a_n$ such that $x^n = \sum_{m=0}^{n} a_m P_m(x)$, and verify that $a_n = \frac{2^n (n!)^2}{(2n)!}$.

7. Let $P_n(x)$ be the Legendre polynomial of degree n. Show that

   (a) $\displaystyle\int_{-1}^{1} x^m P_n(x)\,dx = \begin{cases} 0, & m < n \\ \dfrac{2^{n+1}(n!)^2}{(2n+1)!}, & m = n \end{cases}$

   (b) $\displaystyle\int_{-1}^{1} (1-x^2) P_n'(x) P_n'(x)\,dx = \frac{2n(n+1)}{2n+1}$

   (c) $\displaystyle\int_{-1}^{1} x P_n(x) P_{n-1}(x)\,dx = \frac{2n}{4n^2-1}$

   (d) $\displaystyle\int_{-1}^{1} (x^2-1) P_{n-1}(x) P_n'(x)\,dx = \frac{2n(n+1)}{(2n+1)(2n+3)}$ [Hint: Use Legendre Diff. Eq.]

   (e) $\displaystyle\int_{-1}^{1} x^{n+2k} P_n(x)\,dx = \frac{(n+2k)!\,\Gamma(k+1/2)}{2^n (2k)!\,\Gamma(n+k+3/2)}, \quad k = 0, 1, 2, \ldots$

   (f) $\displaystyle\int_{-1}^{1} P_n(x) P_{n+1}'(x)\,dx = 2, \quad n = 0, 1, 2, \ldots$

   (g) $\displaystyle\int_{-1}^{1} P_n^2(x)\,dx = \frac{2n-1}{2n+1} \int_{-1}^{1} P_{n-1}^2(x)\,dx$ [Hint: Use triple recurrence formula]

   (h) $\displaystyle\int_{-1}^{1} \frac{P_n(x)\,dx}{\sqrt{1-x}} = \frac{2\sqrt{2}}{2n+1}$ [Hint: Use generating function for $P_n(x)$.]

   (i) $\displaystyle\int_{-1}^{1} x P_n'(x) P_n(x)\,dx = \frac{2n}{2n+1}$

   (j) $\displaystyle\int_{-1}^{1} P_m'(x) P_n'(x)\,dx = m(m+1)$ if $n - m \geq 0$ and $n - m$ is even.

   (k) $\displaystyle\int_{-1}^{1} x^m P_n'(x)\,dx = 2$ if $n - m > 0$ and $n - m$ is odd.

   (l) $\displaystyle\int_{-1}^{1} \left[P_n'(x)\right]^2 dx = n(n+1)$

8. Prove that

$$\frac{1-t^2}{(1-2tx+t^2)^{3/2}} = \sum_{n=0}^{\infty} (2n+1) P_n(x)\, t^n.$$

9. Using the value of $\displaystyle\int_0^\pi \frac{d\phi}{a \pm b\cos\phi}$ for $a^2 > b^2$ with $a = 1 - hx$ and $b = h\sqrt{x^2-1}$, prove Laplace's first integral for $P_n(x)$ as

$$P_n(x) = \frac{1}{\pi}\int_0^\pi \left[x \pm \sqrt{x^2-1}\cos\phi\right]^n d\phi, \qquad n \in \mathbb{N}.$$

10. Taking $a = hx - 1$ in the above problem, prove Laplace's second integral for $P_n(x)$ as

$$P_n(x) = \frac{1}{\pi}\int_0^\pi \frac{d\phi}{\left[x \pm \sqrt{x^2-1}\cos\phi\right]^{n+1}}, \qquad n \in \mathbb{N}.$$

11. Consider the associated Legendre equation

$$(1-x^2)y'' - 2xy' + \left[n(n+1) - \frac{m^2}{1-x^2}\right] y = 0, \tag{1}$$

which occurs in quantum physics. Substituting $y(x) = (1-x^2)^{\frac{m}{2}} v(x)$, prove that v satisfies

$$(1-x^2)v'' - 2(m+1)xv' + \big[n(n+1) - m(m+1)\big]v = 0. \tag{2}$$

Prove that $v(x) = \dfrac{d^m P_n(x)}{dx^m}$ satisfies Eq. (2). In this case, $y(x) = (1-x^2)^{\frac{m}{2}} \dfrac{d^m P_n(x)}{dx^m}$ is the bounded solution of Eq. (1), and is called an associated Legendre function.